CSCI 240 -- PA 1

Review Basic C++/Java & OOP

Feel free to discuss and help each other out but does not imply that you can give away your code or your answers! Make sure to read all instructions before attempting this lab. You cannot work with a lab partner for this assignment.

You must use an appropriate provided template from Canvas and output "Author: Your Name(s)" for all your programs. If you are modifying an existing program, use "Modified by: Your Name(s)".

Review basic programming concepts such as selection, repetition, functions/methods, arrays, and classes.

Review important object-oriented programming concepts such as classes, inheritance, and polymorphism.

Explore development environment for C++ or Java and perform the following exercises in either C++ or Java.

Exercise 1 – Write a C++ function or a Java method that accepts an array of int values and returns the count of the number of times when one value is the same as another value (return 0 when they are distinct). Include a driver to test your function/method with at least two test cases: all values are distinct (5, 7, 2, 6, 4) and two or more values are the same (5, 7, 2, 2, 5, 6, 4, 2).

(5, 7, 2, 6, 4) – result is 0
(5, 7, 2, 2, 5, 6, 4, 2) – result is 4 since (5, 5), (2, 2), (2, 2), and (2, 2)

Exercise 2 – Do Exercise 1.10.33 from your C++ textbook or Exercise 1.10.29 from your Java textbook (Birthday Paradox). Use functions/methods in your solution and take advantage of the function/method in exercise 1. Run each experiment 100 times and you need to output three columns like the sample output below (not actual output):

```
People      Count       Percent
   5           0            0%
  10           7            7%
…
 100         100          100%
```

Exercise 3 – Difference Progression

Write a C++ or Java class, **DiffProgression**, that is derived from the Progression class, from the book, to produce a progression where each value is the absolute value of the difference between the previous two values. Include a default constructor that starts with 2 and 200 as the first two values and a two-argument constructor that starts with a specified pair of numbers as the first two values. Try the following test cases and print the first 5 values: create an object using the default constructor and create another object with 15 and 3 as the first two values.

Exercise 4 – Credit Card

Create a new class **MyCreditCard** that derives from **CreditCard** class from textbook to support the following features:

- New credit cards carry annual interest.
- Keep track of all transactions for each credit card.
- Operation `addInterest()` will add the current monthly interest amount to the current balance.
- Operation `printTransactions()` will output all transactions for each credit card.

Set up main to include the following:

1. Create 2 different credit card objects with annual interest of 10%.
2. Run the loop 6 times to simulate 6 months. For each iteration, do the following for each credit card:
    a. Make one or more charges.
    b. Make a payment (can be full or partial amount).
    c. Add monthly interest amount to remaining balance.
    d. Output latest credit card information.
3. Print all transactions for each credit card after the loop. Each transaction should include an amount and the type of transaction such as charge, payment, or interest.

Question 1: List and describe some typical components in a class.

Question 2: What are some good reasons for taking advantage of inheritance instead of creating a new standalone class?

You can earn EC for only one of the options below:

Extra Credit Option 1 (2 points)**:** Perform exercise 1.10.31 from your C++ textbook or exercise 1.10.28 from your Java book. The mistakes should be as realistic as possible and try them with 50 times instead of 100 times. There should be a total of 10 mistakes for all the 50 lines, and it is possible to make more than 1 mistake per line. Use a random letter to simulate a mistake such as “wilt” instead of “will”. Output the number of mistakes at the end of the line if at least one mistake is made. It also makes different mistakes each time you run the program so provide output for 2 different runs to show that.

Extra Credit Option 2 (2 points)**:** Modify exercise 4 to print total charges, average charge, and highest charge for the month when you print each credit card for each month. Do make a few charges for a credit card in a month as a test case. Print all transactions for each credit card to both screen and an output file. You can submit just one version in lieu of exercise 4.

Fill out and turn in the PA submission file for this assignment (save as PDF format).